Conceptual Design and Bench-scale Demonstration of a Hybrid Powertrain for a Small Urban Vehicle

ME 325 – Mechanical Engineering Group Project

Project Proposal Report

Proposed by:

E/21/089	Dewpura A. S.
E/21/091	Dharmapriya B. U. G.
E/21/344	Samarakoon M. M.

Supervised by:

Prof. A. C. Ratnaweera

Department of Mechanical Engineering
Faculty of Engineering
University of Peradeniya

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1 Project Overview

1.1 Introduction

This project focuses on the conceptual design, simulation-based optimization, and bench-scale demonstration of a hybrid powertrain tailored for small urban vehicles, specifically within the L7e category. Utilizing a 1D system modeling approach (Longitudinal Dynamics), the research evaluates the energy flow and efficiency benefits of a series hybrid configuration when subjected to the specific driving conditions of a high-density urban environment.

1.2 Purpose and Need

Urban transportation in developing nations is at a critical juncture. Small vehicles like three-wheelers are essential for mobility but present significant efficiency challenges. Conventional internal combustion engine (ICE) vehicles are designed for steady-state highway cruising, yet they spend the majority of their operational life in erratic urban traffic. The requirement for a hybrid bridge is driven by the need to optimize fuel consumption and reduce localized emissions without the immediate infrastructure burden of full electrification.

2 Scientific Selection of Hybrid Architecture

The selection of a hybrid architecture is a function of the vehicle's intended operational environment. This project provides a scientific justification for the selection of a **Series Hybrid Topology** over Parallel or Series-Parallel alternatives. The decision is grounded in the high transient nature of the Colombo urban drive cycle, specifically targeting the mitigation of energy losses during the idle and acceleration phases identified in local research.

2.1 Quantitative Analysis of the Urban Drive Cycle

Based on the research paper *"Development of a driving cycle for Colombo, Sri Lanka"* (Galgamuwa et al., 2016), the following kinematic parameters were identified for the Colombo metropolitan area:

Table 1: Colombo Drive Cycle Parameters & Architecture Significance

Parameter	**Value**	**Significance for Selection**
Average Speed	20.3 km/h	Indicates low-speed, high-torque demand.
Idle Time	20.5%	Traditional engines waste fuel while stationary.
Acceleration	36.1%	Requires high peak torque; engine rarely hits steady-state.
Deceleration	30.65%	High potential for kinetic energy recovery.
Cruise Mode	12.75%	Minimal opportunity for direct engine-to-wheel drive.

2.2 Engineering Justification for Series Topology

2.2.1 Decoupling of Engine and Road Speed

In a Parallel Hybrid, the ICE is mechanically coupled to the wheels. Given that only 12.75% of the Colombo cycle is spent cruising, a parallel engine would be forced to operate in low-gear, high-transient regions for 87.25% of the time, where thermal efficiency is at its lowest. In a Series Hybrid, the engine is decoupled from the road. This allows the engine to be tuned to operate at its **Optimal Operating Point (OOP)** on the Brake Specific Fuel Consumption (BSFC) map. The engine-generator set maintains the Battery State of Charge (SoC) while the Traction Motor handles the stop-and-go volatility.

2.2.2 Maximizing Regenerative Braking Efficiency

With 30.65% of the drive cycle spent in deceleration, the traction motor acts as a generator to capture energy. The Series architecture is inherently simpler for maximizing regenerative braking in small vehicles because it does not require a complex clutch/transmission interface to disconnect the engine during braking maneuvers.

2.2.3 Mitigation of Idle Losses

The 20.5% idle time is the primary source of emissions in Colombo. In the selected Series architecture, the engine is automatically shut off during idling if the SoC is sufficient, or it runs at a constant, high-efficiency RPM to charge the battery, effectively converting ”wasted idle time” into ”stored energy.”

2.3 Dynamic Tractive Effort Analysis

The architecture was further validated against the Tractive Effort requirement F_{te}, defined by the longitudinal dynamics equation:

$$F_{te} = F_{inertia} + F_{rolling\_resistance} + F_{aerodynamic\_drag} + F_{slope\_component} \tag{1}$$

Where:

- $F_{inertia} = m \cdot a$ (Significant during the 36.1% acceleration phase).
- $F_{rolling\_resistance} = m \cdot g \cdot f_{rr}$ (Constant parasitic loss).
- $F_{aerodynamic\_drag} = \frac{1}{2} \cdot \rho \cdot A \cdot C_d \cdot v^2$ (Negligible at the avg. speed of 20.3 km/h).

The Traction Motor in a Series setup provides peak torque at 0 RPM, making it scientifically superior for overcoming the high $F_{inertia}$ found in the Colombo cycle.

3 Analysis of Existing Systems

3.1 Specifications and Limitations

- **Conventional ICE Small Vehicles:** Rely on 200cc-450cc engines. Limitations include zero energy recovery during braking and peak fuel wastage during the 20.5% idle period identified in local cycles.

- **Pure Electric Vehicles:** While offering zero tailpipe emissions, they are limited by high initial costs and a critical lack of charging infrastructure. For commercial operators, the long downtime for charging results in lost daily revenue.

4 Project Objectives & Methodology

4.1 Objectives

- To analyze urban vehicle requirements and define a suitable hybrid powertrain architecture.

- To size key components (engine-generator, traction motor, and battery) using 1D simulation.

- To develop a Rule-Based Energy Management Strategy (EMS).

- To fabricate a bench-scale demonstrator representing the hybrid powertrain.

- To evaluate system performance in terms of power flow, efficiency, and load response.

4.2 Methodology (1D System Modeling Workflow)

The project utilizes a 1D longitudinal dynamics approach in MATLAB/Simulink. The 7-step engineering workflow derived for this project consists of:

1. **Target Specification:** Defining performance KPIs (Max speed, acceleration).

2. **Detailed Vehicle Specification:** Formulating the control algorithm requirements.

3. **MATLAB Feed:** Integrating vehicle constants into the simulation environment.

4. **Simulation:** Testing the model against the synthesized Colombo Drive Cycle.

5. **Hybrid Control Strategy:** Optimizing the energy management logic for switching modes.

6. **Dynamic Analysis:** Validating the 1D model against calculated tractive efforts.

7. **System Integration:** Mapping 1D simulation results to the physical bench-scale hardware.

5 Project Planning & Timeline

The project follows a 14-week schedule starting from April 27th.

Table 2: 14-Week Project Milestones

Milestone	Focus	Duration
M1	Research, Cycle Analysis & Target Specs	Weeks 1 - 3
M2	1D System Modeling, Component Sizing & Simulation	Weeks 4 - 7
M3	Bench-scale Fabrication & Code Integration	Weeks 8 - 11
M4	Performance Evaluation & Final Reporting	Weeks 12 - 14

6 Impact Analysis

- **Economic:** Targeted 30% reduction in fuel expenditure for urban commercial operators.
- **Social:** Improvement in urban public health through reduced noise and localized air pollution.
- **Environmental:** Enhanced sustainability through kinetic energy recovery and engine downsizing.

7 References

[1] Galgamuwa, U., Perera, L., and Bandara, S. (2016). Development of a driving cycle for Colombo, Sri Lanka. *Journal of Advanced Transportation*.

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[3] Shigley, J. E. (2011). *Mechanical Engineering Design*. McGraw-Hill Education.

[4] Mashadi, B., and Crolla, D. (2011). *Vehicle Powertrain Systems: Integration and Optimization*. John Wiley & Sons.